

# A corpus-based analysis of differences in the use of *very* for adjective amplification among native speakers and learners of English

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This paper analyzes the use of *very* as an adjective amplifier by native speakers and advanced learners of English with diverse language backgrounds based on the *International Corpus of Learner English* (ICLE) and the *Louvain Corpus of Native English Essays* (LOCNESS). The study applies *Multifactorial Prediction and Deviation Analysis Using Regression/Random Forests* (MuPDARF) to find differences between native speakers and advanced learners and evaluate which factors contribute to learners' non-target-like use of *very*. The analysis finds language background and adjective-specific differences in the use of *very* between learners and native speakers. It shows that collocational preferences of specific adjective types are the most important factor, which is interpreted to show that differences between native speakers and learners are predominantly dependent upon the collocational profiles of individual adjective types. This finding supports approaches that focus on teaching collocations and contextualizing word use.

**Keywords:** adjective amplification, L1 background, collocation, MuPDARF

## 1. Introduction

The present study focuses on the use of *very* for adjective amplification by native speakers of English (NS) and advanced learners of English (NNS) with diverse language backgrounds. Adjective amplification is particularly relevant from the perspective of learners of English because this frequent linguistic strategy plays a crucial role in the social and emotional expression of speakers (see Labov, 1985; Partington, 1993; Peters, 1992; Ito & Tagliamonte, 2003: 258). As such, adjective amplifiers represent key elements in expressing stance (Fallas Escobar & Chaves

Fernández, 2017) and they allow both NS and NNS to indicate social affiliations within a given speech community, thus demonstrating social and pragmatic relevance (Hyland, 2005:178–179). However, appropriately acquiring the intricate social and linguistic multifunctionality of individual amplifier variants represents a serious challenge, even for advanced learners. This is because amplifier systems are a site of “fevered invention” and a domain that is particularly prone to language change (Bolinger, 1972:18; Quirk, Greenbaum, Leech, & Svartvik, 1985:590).

This study specifically focuses on the use of *very* by NS and NNS as it is not only traditionally the most frequent adjective amplifier across modalities, but it remains the dominant adjective amplifier in writing, which is of particular importance in the educational context. *Very* is an upscaling adjective amplifier (cf. Quirk et al., 1985:589–590) (see [1] and [2]), and its use is analyzed against variants that could have served as amplifying alternatives such as *really*, *so*, or *awfully*. Therefore, the present study leaves aside downscaling intensifiers (including approximators such as *almost*, compromisers such as *more or less*, diminishers such as *partly*, and minimizers such as *hardly*) because such forms, while also classified as intensifiers (cf. Quirk et al. 1985:589–590), perform a semantically different function, namely downtoning, rather than amplifying.

- (1) Adjective amplification with *very* in predicative contexts
  - a. [...] their parents objected to the marriage because they were *very poor*.<sup>1</sup>  
(ICLE-BG-SUN-0003.1)
  - b. This fact can be *very beneficial* for Poland. (ICLE-PO-POZ-0041.5)
  - c. [...] competition on job markets is *very high* these days and there exists a real threat of unemployment [...]. (ICLE-RU-MOS-0025.4)
- (2) Adjective amplification with *very* in attributive contexts
  - a. [...] Europe has to become a *very strong* power just like the American and the Japanese “Giants”. (ICLE-FR-UCL-0066.1)
  - b. “But in our pubs,” my informants proudly say, “it is a *very different* kettle of fish.” (ICLE-DN-NIJ-0002.2)
  - c. There is always somebody who will think differently from you, as it is a *very personal* question. (ICLE-SW-LND-0008.9)

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1. In the examples, the code ICLE stands for the corpus (ICLE=International Corpus of Learner English). The next two capital letters represent a country or language code (BG=Bulgaria, CZ=Czech Republic, DB=Dutch [Belgium], DN=Dutch [Netherlands], FI=Finland, FR=France, GE=Germany, IT=Italia, PO=Poland, RU=Russia, SP=Spain, SW=Swedish). The next three capital letters refer to the city in which the data was collected, for example LND for Lund, NIJ for Nijmegen, POZ for Poznan, etc. The numbers at the end of the code identify the essay and the author of the essay.

Previous research has produced a wealth of insights into adjective amplification, adding to our understanding of differences between NS and NNS in argumentative texts (see Section 2). The most consistent finding produced by these studies is an overuse of general intensifiers, most prominently *very*, by NNS (Granger, 1998a; Lorenz, 1999: 215; Philip, 2008). However, Gries (2018) points out that the concepts of over- and underuse can be misleading if not used in the sense of significant deviations from what would be expected in NS output.<sup>2</sup> In addition, Gries (2018) argues that only statistics that control for confounding factors allow us to accurately unearth quantitative differences between NNS and NS. The present study follows Gries' (2018) suggestion and analyzes the use of *very* by NS and advanced NNS with diverse language backgrounds by applying a *Multifactorial Prediction and Deviation Analysis Using Regression/Random Forests* approach (MuPDARF, see Section 3.2 for more details about this method) to the *International Corpus of Learner English* (ICLE; Granger, Dagneaux, & Meunier, 2002; Granger, 1993) and the *Louvain Corpus of Native English Essays* (LOCNESS; Granger, 1998b).

## 2. Previous research

Adjective amplification has received a substantial amount of attention (e.g. Lorenz, 1998, 1999, 2002; Philip, 2008; Granger, 1998a; Hendriks, Van Goethem, & Wulff, 2019). One of the most consistent findings of previous research relates to overuse of *very* and *really* in argumentative texts produced by intermediate to advanced NNS (see Granger, 1998a; Lorenz, 1999: 215). Philip (2008), for instance, found that *very* and *really* are used to the "virtual exclusion of any other adverb" (Philip, 2008: 1308). Similarly, Lorenz (1999) found that German learners of English overuse the all-purpose intensifiers *very* and *really*, which they use in a way more representative of speech than argumentative texts (1999: 198, 215). Lorenz (1998: 62–63) argues that the tendency among German NNS to amplify more frequently represents a misapplication due to overall writing strategies grounded in an NNS's desire for expressiveness. The latter point echoes analyses of the expression of stance in argumentative texts by NNS which found that the use of boosters (amplifiers), together with other stance-taking strategies, enable students to develop a stronger voice (Fallas Escobar & Chaves Fernández, 2017: 118). Hinkel (2003) has argued in a similar vein, stating that the patterning

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2. It is crucial to note that the terms "deviation", "difference", "divergence", or "non-target-like", as well as the notion of what is expected, are not used to indicate deficiencies on the part of NNS but to refer to statistical or quantitative differences. As such, these phrases are used descriptively and neither teleologically nor in a sense that implies value-judgements.

of adverbs in data coming from NNS with Chinese, Japanese, Korean, and Indonesian language backgrounds mirrored the frequencies in conversational style. Hinkel (2003) interpreted this to indicate that learners used a colloquial style rather than a more appropriate formal style due to a limited lexical repertoire (2003:1058).

Furthermore, both Lorenz (1998, 1999) and Granger (1998a) found that NNS show a higher degree of variability in amplifier-adjective collocations. Among the German NNS studied by Lorenz (1998) based on a subsection of the ICLE, this trend is captured as a lower degree of cohesion among amplifier-adjective bigrams. Granger (1998a) used both corpus data as well as word combination tasks to tap into intuitional differences concerning amplifier-adjective collocations between NS and French learners of English. The word combination task indicated that learners demonstrated significantly more variation in what they deemed acceptable pairs compared with NS. This, according to Granger (1998a:152), suggests that the sense of collocational salience was missing or misguided among the learners. Similarly, Pérez-Paredes and Díez-Bedmar (2012) found that young Spanish NNS only used a limited set of amplifiers, concluding that “it would take several years [for very young learners of English] to start using a more varied set of intensifying adverbs other than *very*” (2012:117).

Several studies also reported that differences between NS and NNS, which waned as NNS became more proficient. Results provided by Edmonds and Gudmestad (2014), who analyze intensifying adverbials in argumentative texts, indicated that advanced NNS of English performed like English NS while less advanced NNS differed substantially from both advanced NNS and NS. This trend is corroborated by Forsberg (2010) who focused on the use of formulaic sequences among learners of French and also found that differences between learners and native speakers of French waned once learners became more proficient.

Similarly, Hendrikx et al. (2019:80–81) used covarying analyses to investigate the impact of *Content and Language Integrated Learning* (CLIL) on the acquisition of intensifying constructions (including amplifier-adjective bigrams) among native speakers and learners of French, Dutch, and English. They found that while learners amplified significantly less often than native speakers, more advanced learners approximate the rate of amplification exhibited by native speakers (though this effect was more apparent for learners of English compared to learners of French and Dutch). In addition, students in CLIL produced more target-like intensification compared with students in non-CLIL. Target-like use of intensification was shown, e.g., by “less overuse of all-round intensifiers and collocations containing advanced vocabulary as well as informal intensifying language” (Hendrikx et al., 2019:97).

Despite providing relevant insights into the use of adjective amplifiers by NNS, the variety of L1 learner populations investigated is rather limited, focusing predominantly on French, German, and Dutch learners; only Hinkel (2003) focused on learners with non-Indo-European language backgrounds. Furthermore, all comparative studies that have investigated adjective amplification among NS and NNS have relied on mono- or bivariate statistics and were thus not able to control for confounding factors. Given the existing research, at least two issues remain unresolved with respect to adjective amplification among NNS:

1. While several studies have claimed that *very* is overused by NNS (e.g. Granger, 1998a; Lorenz, 1999: 215; Philip, 2008: 1308), they did not use multivariate methods which would have allowed to control for confounds. Thus, the first research question addressed in this paper is whether a multifactorial approach that controls for confounds can confirm that *very* is overused by NNS.
2. Given previous research, it remains unclear if the overuse of *very* in argumentative texts by learners of English is mainly adjective-specific or dependent on the language background of NNS. Accordingly, the second research question addressed in this study is what factors correlate with non-target-like use of *very* for adjective amplification.

### 3. Data and methodology

The present paper combines Learner Corpus Research (LCR) with recently introduced multivariate statistical methods to identify systematic differences between NS and NNS in the use of *very* for adjective amplification and to investigate which factors contribute to non-target-like performance. The following section consists of two subsections which describe the data and the methodology that have been applied to the data in this study.

#### 3.1 Data sources and processing

This study is based on two data sources. The first data source is the first version of the ICLE. The corpus consists of argumentative writing by intermediate to advanced learners of English with Bulgarian, Czech, Dutch, Finnish, Flemish, French, German, Italian, Polish, Russian, Spanish, and Swedish language backgrounds.<sup>3</sup>

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3. Flemish (Belgian Dutch) and Dutch (as spoken in the Netherlands) are not collapsed in this study following the distinction between these two varieties by the corpus compilers of ICLE.

The second data source is the LOCNESS which was specifically designed to provide data comparable to the ICLE. The LOCNESS contains English NS essays and consists of British pupils' A level essays (60,209 words), British university students' essays (95,695 words), and American university students' essays (168,400 words). Although produced by younger pupils than university students, A level essays of British pupils were included to achieve a corpus size that is sufficient to arrive at meaningful statistical results.

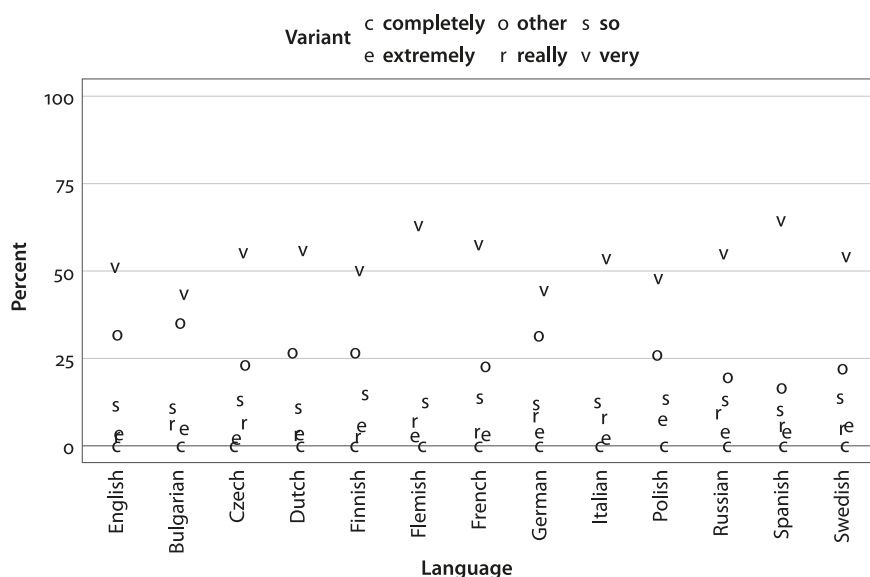
The data processing of the corpus data was done in the programming environment R (R Core Team, 2019). Before further processing, both data sets were combined into a single data frame where each row contained a single essay or text, which was then cleaned and standardized by removing, for instance, meta-data such as file identifiers. The data were then part-of-speech tagged by implementing a maximum entropy tagger provided in the *openNLP* package (Hornik, 2016). The reliability of the automated part-of-speech tagging was evaluated based on a random sample of 200 sentences (2,100 elements, 1,795 words) from the data. The sample was manually tagged to assess the accuracy of the automated tagging, which showed that the automated tagging was correct in 99.4 percent of cases for non-base adjectives (e.g. *bigger*, *biggest*, *more beautiful*, *most beautiful*) and in 94.6 percent of cases where the word was a base adjective (e.g. *big*, *beautiful*). The relative frequency of adjectives (percent of each adjective out of all adjectives) was calculated separately for each language and subsequently logged and then scaled to obtain a measure which can be used to control for frequency effects. Next, all adjectives (tagged JJ) were extracted and the adjective type as well as its syntactic function (predicative versus attributive) were determined. To assess the accuracy of the assignment of syntactic patterns, a random sample of 200 adjectives was drawn. The automated classification was compared with a manual classification which showed that 97.1 percent of attributive adjectives and 88.9 percent of predicative adjectives were classified correctly.

In addition, it was determined whether the adjective was amplified and which lexical form served as an amplifier. The classification of whether an adjective was amplified relied on a list of amplifiers<sup>4</sup> and was also checked manually in order

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4. The list of amplifiers contained the following elements: *absolutely*, *awful*, *awfully*, *bloody*, *certainly*, *clearly*, *completely*, *crazy*, *dead*, *decidedly*, *definitely*, *distinctly*, *dreadfully*, *enormously*, *entirely*, *exceedingly*, *exceptionally*, *excruciatingly*, *extraordinarily*, *extremely*, *fiercely*, *firmly*, *frightfully*, *fucking*, *fully*, *genuinely*, *greatly*, *grossly*, *heavily*, *highly*, *hopelessly*, *incredibly*, *mad*, *mega*, *mighty*, *obviously*, *openly*, *overwhelmingly*, *particularly*, *perfectly*, *plenty*, *positively*, *precisely*, *pretty*, *profoundly*, *purely*, *real*, *really*, *remarkably*, *seriously*, *significant*, *significantly*, *so*, *specifically*, *strikingly*, *strongly*, *substantially*, *super*, *surely*, *terribly*, *terrifically*, *total*, *totally*, *traditionally*, *true*, *truly*, *ultra*, *utterly*, *very*, *viciously*, *wholly*, *wicked*, and *wildly*.

to avoid missing instances of amplification or incorrect classification. It was then tested if the same amplifier type had occurred within a span of up to three previous pre-adjectival slots to evaluate potential persistence or priming effects (cf. Tulving & Schacter, 1990:301; Szmrecsanyi, 2005:113; also Szmrecsanyi, 2006). Priming in the present study refers to instances of production priming (cf. Szmrecsanyi, 2005:113) and can therefore be defined as the re-use of linguistic material that was used in the preceding discourse (cf. Tulving & Schacter, 1990:301). Non-amplified instances of adjectives, negated adjectives, misclassified items, as well as comparative and superlative forms were then removed from the analysis. Adjectives that were never amplified, or which were not amplified by at least two different amplifier types, were also removed from the analysis to avoid including lexicalized forms such as the *right honourable*. Given that random forest models can only handle variables with a maximum of 52 levels, all remaining adjectives that occurred less than 30 times were collapsed into a single category ('other'), leaving the data with 39 adjective types (in contrast to adjective tokens). Due to the low frequency of amplifiers other than *very* (see Figure 1), all remaining amplifier types were collapsed into a single category ('other').



**Figure 1.** Percentages of amplifier variants by language with variants other than *completely*, *extremely*, *so*, *really*, and *very* collapsed

Figure 1 shows that *very* is the most frequent adjective amplifier in the data, irrespective of language background and that there are notable frequency differences in the use of amplifier variants across language backgrounds although the overall patterning of amplifier variants remains relatively consistent.

In a preliminary analysis in which semi-frequent amplifier variants such as *really*, *so*, or *pretty* were not collapsed, the random forest models had very low explanatory power and only predicted the use of *very* or *other* amplifiers while never predicting the use of semi-frequent variants; this led to the decision to collapse all amplifier variants, with the exclusion of *very*, into a single category.

In addition to annotating the language background (*language*) of speakers, the study applied a sentiment analysis (Jockers, 2017) to the adjectives, as amplifier choice has been shown to be affected by the emotionality of the amplified adjective. The sentiment analysis used in the current study relies on the *Word-Emotion Association Lexicon* (Mohammad & Turney, 2013), which comprises 10,170 terms based on 38,726 ratings from 2,216 raters. The emotion coding of this lexicon is based on ratings gathered through the crowd-sourced *Amazon Mechanical Turk* service. In this Turk survey, raters were asked a sequence of questions relating to whether a given word was associated with one of eight emotions (ANGER, ANTICIPATION, DISGUST, FEAR, JOY, SADNESS, SURPRISE, TRUST). Each term was rated at least five times and for 85 percent of words, four or more raters provided identical ratings. According to the ratings, words like *dark* or *tragic* are more readily associated with SADNESS and words such as *happy* or *beautiful* are associated with JOY, while words like *cruel* or *outraged* are associated with ANGER. In the present study, adjectives associated with ANGER, DISGUST, FEAR, or SADNESS are coded as negative (e.g. *sad*, *angry*) while adjectives associated with ANTICIPATION, JOY, SURPRISE, or TRUST are coded as positive (e.g. *happy*, *nice*). Adjectives that are not associated with any emotional state are coded as non-emotional (e.g. *rusty* or *flat*).

In addition, adjectives were classified semantically based on a simplified version of the classification provided by Dixon (1977, 2004; cf. also D'Arcy, 2015; Tagliamonte, 2008; Tagliamonte & Roberts, 2005). The reason for simplifying Dixon's (1977, 2004) categorization relates to the uneven distribution of category members. Because the data contained hardly any age and color terms, such adjective types as well as adjectives that could not be categorized (e.g. *familiar*, *genuine*, or *inadequate*) were assigned the label *Other*. As a result, the semantic classification consists of five levels (Difficulty, Dimension, HumanPropensity, Other, PhysicalProperty, and Value).



Finally, each adjective was assigned a gradability score. Gradability represents a semantic property of adjectives (cf. Quirk et al., 1985) that encodes the degree of a characteristic (Biber, Johansson, Leech, Conrad, & Finegan, 2007: 521) and thus refers to a comparison to a baseline. In contrast to non- or ungradable adjectives, gradable adjectives can occur in contexts that denote comparative or superlative states. In such contexts, gradable adjectives either take morphological marking with <-er> or <-est> or periphrastic marking with 'more' or 'most', while the base forms of gradable adjectives in non-comparative contexts do not have any special marking. Some adjectives rarely form a comparative or superlative, for example *\*more dental* or *\*?most motionless*. While amplifiers are typically not used with non- or ungradable adjectives, such adjectives can be amplified for pragmatic purposes such as emphasis. This suggests that gradability reflects a quantitative rather than a qualitative property of adjectives. To accommodate the fact that gradability is not categorical, gradability is operationalized as the logit of the probability of an adjective being used in a comparative context versus the overall rate of comparison among adjectives. Since the data used in the present study is insufficient to serve as the basis for such a calculation, the gradability score was calculated based on the *British National Corpus* (BNC; 2007). In the present study, a gradability score of 0 represents a neutral state where an adjective occurs at an average rate in comparative contexts, while negative values indicate a tendency towards non-gradability (the adjective occurs less often in comparative contexts than the average adjective) and positive scores indicate a tendency towards gradability (the adjective occurs more frequently in comparative contexts than adjectives do on average).

It should be noted that the variable context of this present study reflects the decision point where a speaker decides which amplifier to use rather than whether to amplify. The variable context thus encompasses only amplified adjectives while leaving out zero contexts, i.e. contexts where the speaker could have amplified an adjective but did not do so. After the data processing, the study arrived at the data set summarized in Table 1.

Table 1 shows that the final data set includes 6,712 amplified adjectives produced by 2,893 NNS. The data contains 3,564 instances of *very* which amounts to 53.1 percent of overall amplification. Numeric variables are not included in Table 1 as it is not useful to determine the number of learners, instances of *very*, and adjective slots for each level of numeric variables.

The variables that were entered into the Random Forest models (during the initial steps of the MuPDARF) are summarized in Table 2.

**Table 1.** Summary of the final data set with respect to the number of learners, adjective slots, instances of *very*, and the percentage of *very* across each variable level

| Variables        | Levels            | Learners | Very<br>(N) | Amplified<br>Adjectives (N) | Percent<br>(very) |
|------------------|-------------------|----------|-------------|-----------------------------|-------------------|
| Language         | English           | 191      | 191         | 377                         | 50.7              |
|                  | Bulgarian         | 205      | 196         | 454                         | 43.2              |
|                  | Czech             | 206      | 321         | 579                         | 55.4              |
|                  | Dutch             | 109      | 236         | 422                         | 55.9              |
|                  | Finnish           | 175      | 228         | 455                         | 50.1              |
|                  | Flemish           | 130      | 170         | 270                         | 63.0              |
|                  | French            | 275      | 300         | 524                         | 57.3              |
|                  | German            | 280      | 257         | 581                         | 44.2              |
|                  | Italian           | 247      | 254         | 476                         | 53.4              |
|                  | Polish            | 283      | 316         | 664                         | 47.6              |
|                  | Russian           | 226      | 363         | 658                         | 55.2              |
|                  | Spanish           | 209      | 354         | 551                         | 64.2              |
|                  | Swedish           | 357      | 378         | 701                         | 53.9              |
| Emotionality     | NegativeEmotional | 860      | 461         | 1083                        | 42.6              |
|                  | NonEmotional      | 2235     | 2070        | 3867                        | 53.5              |
|                  | PositiveEmotional | 1250     | 1033        | 1762                        | 58.6              |
| Priming          | NoPrime           | 2891     | 3440        | 6553                        | 52.5              |
|                  | Prime             | 144      | 124         | 159                         | 78.0              |
| SemanticCategory | Difficulty        | 412      | 311         | 460                         | 67.6              |
|                  | Dimension         | 550      | 443         | 634                         | 69.9              |
|                  | HumanPropensity   | 648      | 374         | 761                         | 49.1              |
|                  | Other             | 1,662    | 1,111       | 2558                        | 43.4              |
|                  | PhysicalProperty  | 570      | 339         | 671                         | 50.5              |
|                  | Value             | 1,181    | 986         | 1628                        | 60.6              |
| Total            |                   | 2,893    | 3,564       | 6,712                       | 53.1              |

**Table 2.** Overview of variables evaluated in the present study

| Variable          | Scale       | Levels/parameters       |          | Description                                                                |
|-------------------|-------------|-------------------------|----------|----------------------------------------------------------------------------|
| Dependent         |             |                         |          |                                                                            |
| very              | binary      | yes; no                 |          | Presence/absence of very in pre-adjectival position                        |
| Independent       |             |                         |          |                                                                            |
| Learner           | categorical | 2,893 learners          |          | Learner that produced the adjective                                        |
| Adjective         | categorical | aware:young; * 39 types |          | Adjective that is being amplified                                          |
| Language          | categorical | Bulgarian;              | Czech;   | Language background of the learner                                         |
|                   |             | Dutch;                  | English; |                                                                            |
|                   |             | Finnish;                | Flemish; |                                                                            |
|                   |             | French;                 | German;  |                                                                            |
|                   |             | Italian;                | Polish;  |                                                                            |
|                   |             | Russian;                | Spanish; |                                                                            |
|                   |             | Swedish                 |          |                                                                            |
| Priming           | binary      | Prime; NoPrime          |          | Presence/absence of the same amplifier in up to three pre-adjectival slots |
| Gradability       | numeric     | Min.:                   | −3.66;   | Score representing the tendency of an                                      |
|                   |             | Median:                 | −0.37;   | adjective to occur in comparative or                                       |
|                   |             | Mean:                   | −0.84;   | superlative contexts                                                       |
|                   |             | Max.:                   |          |                                                                            |
|                   |             | 2.73                    |          |                                                                            |
| RelativeFrequency | numeric     | Min.:                   | −4.87;   | Relative frequency of adjective by language                                |
|                   |             | Median:                 | −1.16;   | background                                                                 |
|                   |             | Mean:                   | −1.39;   |                                                                            |
|                   |             | Max.:                   |          |                                                                            |
|                   |             | 1.42                    |          |                                                                            |
| Emotionality      | categorical | NegativeEmotional;      |          | Association with positive or negative core                                 |
|                   |             | NonEmotional;           |          | emotions                                                                   |
|                   |             | PositiveEmotional       |          |                                                                            |
| Function          | binary      | Attributive;            |          | Syntactic function/context of the adjective                                |
|                   |             | Predicative             |          |                                                                            |
| SemanticCategory  | categorical | Difficulty;             |          | Semantic category of the adjective                                         |
|                   |             | Dimension;              |          |                                                                            |
|                   |             | HumanPropensity;        |          |                                                                            |
|                   |             | PhysicalProperty;       |          |                                                                            |
|                   |             | Value; Other            |          |                                                                            |

\* See the x-axis tick labels of Figure 3 for the entire list of adjective types used in this study.

### 3.2 Statistical methods

The rise of LCR in the past three decades or so has enabled us to analyze L1-specific difficulties among learners from diverse language backgrounds. In addition, increasingly sophisticated statistical modeling allows us to analyze factors that condition learners' choices associated with cognitive mechanisms that underlie target-like speech production (see, e.g., Gries, 2018; Gries & Adelman, 2014; Gries & Deshors, 2014; Wulff & Gries, 2019a; Wulff & Gries, 2019b). The present study uses a recently introduced statistical method, *Multifactorial Prediction and Deviation Analysis Using Regression/Random Forests* (MuPDARF; Gries & Deshors, 2014; Heller, Bernaisch, & Gries, 2017), to provide a fine-grained understanding of which factors contribute to non-target-like choices among NNS from diverse language backgrounds with respect to the use of *very* as an adjective amplifier.

In contrast to mono- or bivariate methods, MuPDARF enables us to investigate the difference between the choices of NS and NNS in a very detailed manner as it controls for confounding factors by simultaneously incorporating multiple variables and interactions between variables. In the present case, MuPDARF allows us to investigate which factors correlate with NNS making non-target-like choices with respect to adjective amplification using *very* and how strongly these factors affect these choices. MuPDARF consists of the following steps:

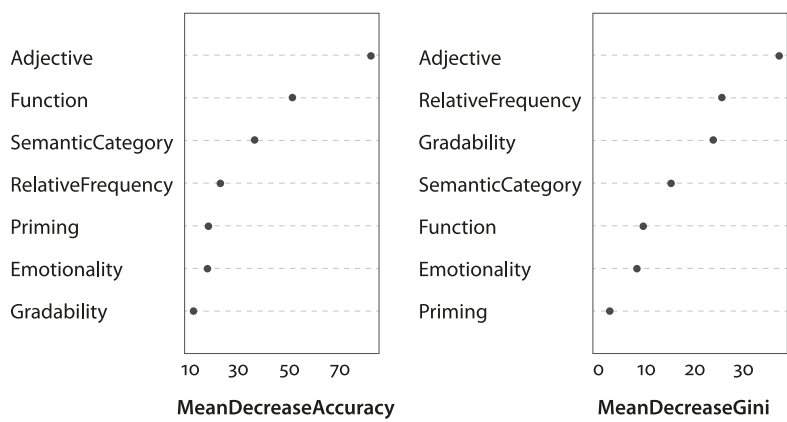
1. Fit a random forest model to the NS data to determine which factors contribute to the decision of an NS and to what extent.
2. Apply the random forest model obtained from the NS data to the NNS data to predict which choice an NS would have made given the same conditions.
3. Save the predicted choices in a vector and compare the choices actually made by the NNS to the predicted choices. This part of the analysis allows us to identify with which adjectives *very* is over- or underused given the specific language backgrounds of the learners. This interim analysis uses Fisher's Exact tests with Benjamini-Hochberg corrections for multiple testing (see Field, Miles, & Field, 2012: 429–431).
4. Then, a second vector is created which shows where the observed and the predicted choices are identical and where they differ. Thereby, an additional variable is added to the data which serves as the dependent variable of a mixed-effects binomial logistic regression. The newly created variable reflects whether the observed behavior of a given NNS differs from the prediction of what an NS would have done in the respective context based on the random forest model. If the observed choice of the NNS agrees with the prediction, the instance is coded as 0 while it is coded as 1 if the observed choice and the prediction differ.

5. An inferential analysis of that second vector is carried out by means of a mixed-effects regression model to identify which factors contribute to NNS making non-target-like choices.

The regression analysis that is performed as a final step of MuPDARF determines which independent variables correlate with the dependent variable (non-target-like choice) and provides information about the direction of the effects and whether the variables impact non-target-like behavior as main effects or as part of interactions. The regression analysis included both adjective type and speaker (the students who produced the amplified adjective) as random effects. Other random effect structures were explored, but resulted in models with higher Akaike Information Criterion (AIC) values, indicating a suboptimal model fit. Model fitting was done by following a forward stepwise selection procedure by which predictors (variables and interactions between variables) are consecutively added to a model. Predictors were retained if they significantly improved model fit and if their inclusion did not result in unacceptable multicollinearity. Multicollinearity was measured using variance inflation factors (VIFs) with a cut-off value of 3 for main effects (see Zuur, Ieno, & Elphick, 2010) and 100 for interactions, as higher collinearity is to be expected for interactions between predictors. In cases where the VIFs were unacceptable, model fitting continued without that predictor. This model fitting procedure aimed to find the final minimal adequate regression model that can explain the most variance with a minimum number of predictors.

#### 4. Results

The initial random forest analysis used the NS data with *very* as the dependent variable and included all other variables with the exception of language and learner as predictors. The model was fit with 3,000 trees while considering two randomly selected variables at each split. Although the out-of-bag error rate of the random forest model was relatively high with 31.3 percent, it still achieved an accuracy of 84.6 percent. The model performed significantly better than a baseline model which only predicted 50.66 percent of the data correctly and thus improved the prediction accuracy by 67.0 percent compared to a baseline model. The explanatory capacity and fit of the model are excellent (Harrell's  $C = .847$ , Somers'  $D_{xy} = .694$ ). An overview of the variable importance measures for the random forest model fit to the NS data is provided in Figure 2.



**Figure 2.** Variable importance plots for the random forest fit to the NS data with use of *very* as dependent variable

We now turn to the results of the application of the random forest model to the NNS data with *very* as the dependent variable. The random forest model achieved 61.7 percent accuracy, which represents a significant improvement by a factor of 15.8 percent compared to a baseline accuracy of 53.24 percent. However, the explanatory capacity and the fit of this second random forest model are poor (Harrell’s  $C = .619$ , Somers’  $D_{xy} = .239$ ).

The next step of the analysis used Fisher’s Exact tests with Benjamini-Hochberg corrections for multiple testing to see if *very* is over- or underused with specific adjectives by NNS based on the results of the random forest model that was fit to the NS data and then applied to the NNS data. The results of this analysis are provided in Table 3.

**Table 3.** Results of significant differences in the co-occurrences of *very* with specific adjective types by language background

| Language | Adjective | NAdjective | Very <sub>NNS</sub> | Very <sub>NS</sub> | φ   | Type     | Sig. Benjamini-Hochberg |
|----------|-----------|------------|---------------------|--------------------|-----|----------|-------------------------|
| Italian  | different | 37         | 15                  | 0                  | 0.5 | Overuse  | $p < .01$               |
| Spanish  | different | 31         | 18                  | 0                  | 0.6 | Overuse  | $p < .001$              |
| Swedish  | different | 30         | 15                  | 0                  | 0.6 | Overuse  | $p < .01$               |
| Polish   | difficult | 35         | 16                  | 35                 | 0.6 | Underuse | $p < .001$              |
| Polish   | important | 44         | 28                  | 44                 | 0.5 | Underuse | $p < .01$               |
| Swedish  | important | 59         | 42                  | 59                 | 0.4 | Underuse | $p < .01$               |
| Swedish  | new       | 12         | 0                   | 12                 | 1.0 | Underuse | $p < .001$              |

The results presented in Table 3 confirm that NNS deviate from NS only with respect to a limited set of adjectives (*different*, *difficult*, *important*, and *new*) and that these differences only emerge across a limited set of language backgrounds (Spanish, Polish, Swedish, Italian). Additionally, the results show that *very* is overused only with *different* (by NNS with Italian, Spanish, and Swedish language backgrounds) while it is underused with *difficult* and *important* by Polish NNS and with *important* and *new* by Swedish NNS. We now turn to the analysis that aimed to determine which factors correlate with non-target-like adjective amplification.

The model fitting process which investigated whether any combination of the independent variables had an effect on non-target-like use of *very* arrived at a final minimal model that performed significantly better than a baseline model (L.R.  $\chi^2$ : 500.69,  $df=33$ ,  $p<.001$ ) despite a suboptimal fit (Harrell's  $C=.681$ , Somers'  $D_{xy}=.361$ ,  $R^2_{\text{marginal}} 0.042$ ,  $R^2_{\text{conditional}} 0.315$ ). As the suboptimal, yet significant, fit gives reasons for concern and requires additional attention, it is addressed further in the discussion section. This final model reported two significant main effects: the language background of speakers and the semantic category of the adjectives. In addition, the model reported three significant interactions:

- a two-way interaction between syntactic context and the relative frequency of adjectives
- a two-way interaction between the syntactic context and priming
- a three-way interaction between syntactic context, emotionality, and gradability of adjectives

The main effects of predictors that are part of significant interactions are not interpreted here following Field et al. (2012: 624).

The individual factors that emerge as significant in the regression model are visualized and briefly summarized in the following paragraphs. We begin with the intercept adjustments (random effects) of the regression model. Figure 3 shows that the intercept adjustments differ substantially across adjective types. This indicates that the proportion of non-target-like use of *very* by NNS varies substantially across adjective types, with the most severe problems being associated with the adjectives *new*, *dependent*, and *likely*. By contrast, the negative adjustments that are necessary for *little* and *long* suggest that NNS use *very* in a target-like manner when amplifying these adjectives.

Given the high number of individual learners, the adjustments by learner are not visualized here but rather summarized numerically. The adjustments for the individual learner range between  $-0.084$  to  $0.084$ , which is substantially smaller compared to the adjustments for adjectives, which range from  $-1.705$  to  $3.149$  (see Figure 3). The low variability among NNS, the relatively high value of  $R^2_{\text{conditional}}$  ( $0.315$ ) (see Table 4 in the Appendix), and the comparatively low  $R^2_{\text{marginal}}$  value

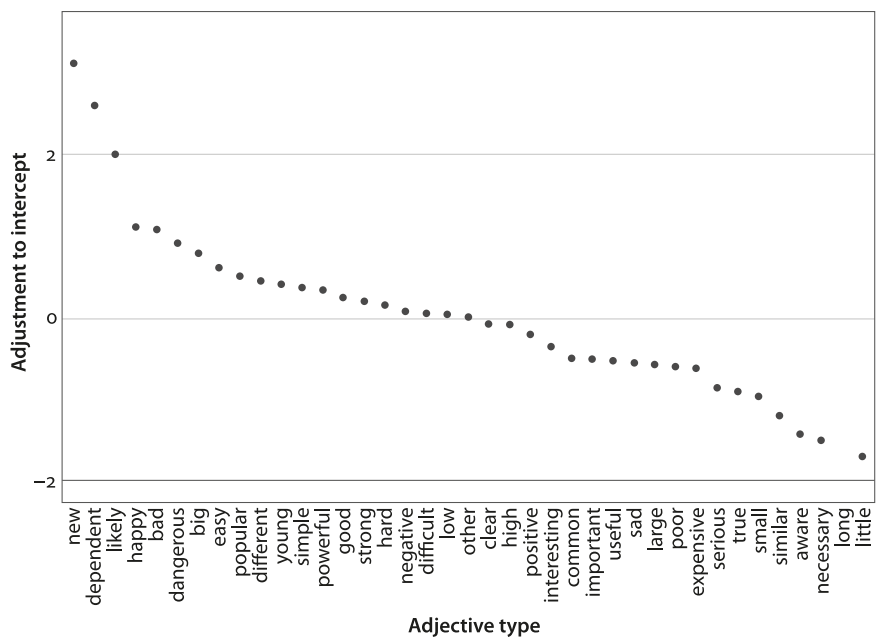


Figure 3. Adjustment by adjective type

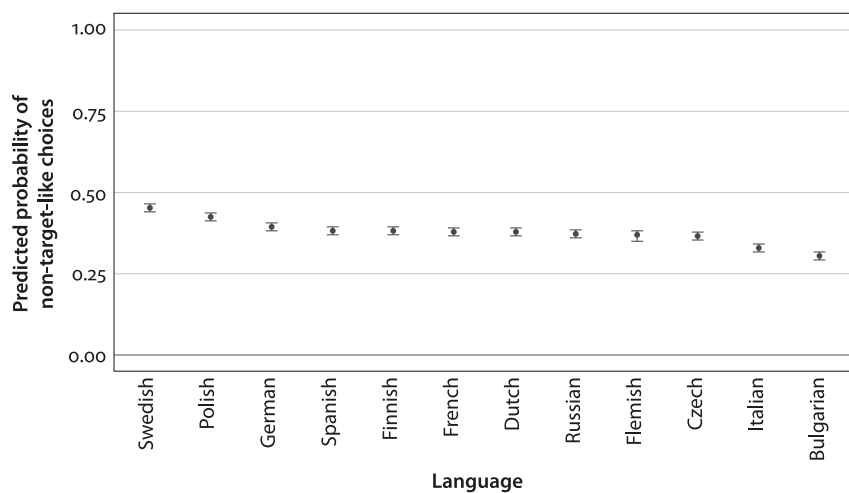
(0.042) indicate that the most important conditioning factor for non-target-like use of *very* in the present study is adjective type.

With respect to significant fixed effects, the model output reported that the language background as well as the semantic category of adjectives impact the likelihood of NNS making non-target-like choices. Figure 4 shows the predicted probability of non-target-like adjective amplification by language background. The ordering of the language backgrounds ranges from highest to lowest effect, with L1 Swedish speakers demonstrating the highest probability of amplifying adjectives in a non-target-like manner and speakers with a Bulgarian language background amplifying most like NS. As such, speakers with Swedish and Polish language backgrounds differ most from NS while speakers with Bulgarian and Italian language backgrounds make the most target-like choices.

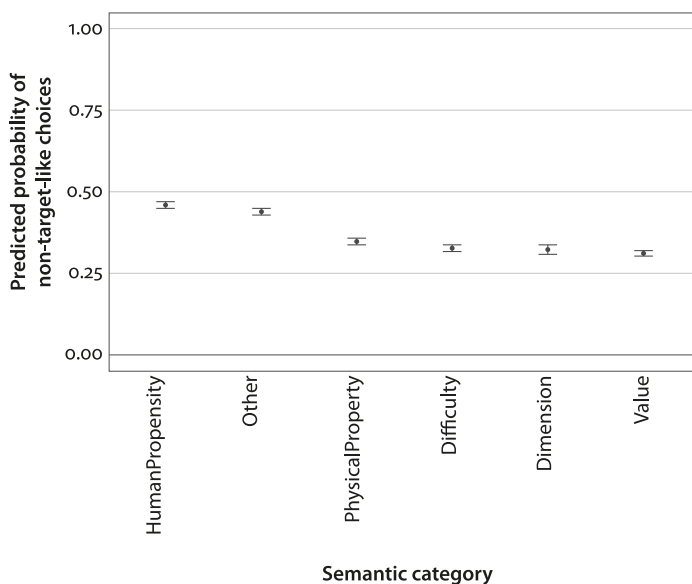
Figure 5 shows the predicted probability of NNS making non-target-like choices with respect to the use of *very* by the semantic category of the adjective.

Figure 5 shows that NNS are more likely to make non-target-like choices when the adjectives denote human propensity, which encompasses adjectives such as *aware*, *happy*, and *sad*, or when the adjectives could not be categorized properly (e.g. *common*, *dependent*, *likely*). Less problematic appear to be adjectives that fall into the categories of value (e.g. *good*, *bad*, *expensive*, or *dangerous*), dimension (e.g. *big*, *high*, or *little*), difficulty (e.g. *difficult*, *easy*, or *simple*), and physical property (e.g. *clear*, *different*, *hard*, or *strong*).





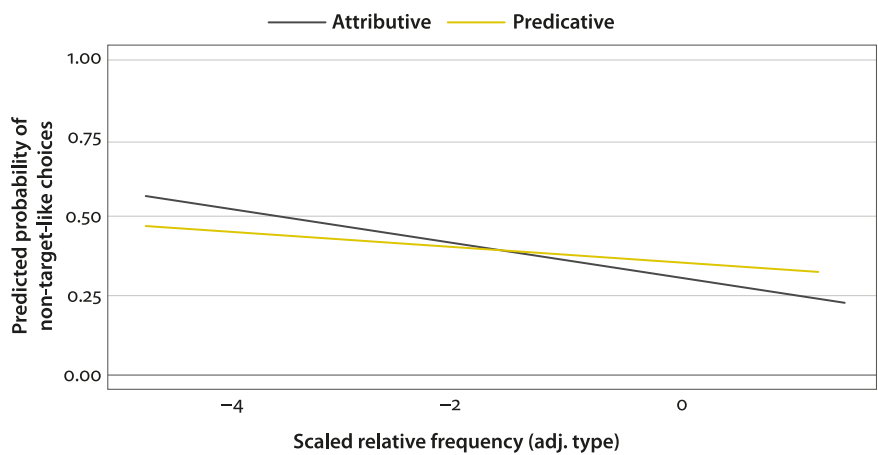
**Figure 4.** Predicted probability of non-target-like choices in adjective amplification with *very* by language background



**Figure 5.** Predicted probability of non-target-like choices in adjective amplification with *very* by semantic category of the adjective

The model output reports two significant two-way interactions: an interaction between syntactic context and adjective frequency as well as an interaction between priming and the syntactic contexts.

According to the regression output, NNS make significantly more non-target-like choices in attributive contexts when amplifying low frequency adjectives, while making more non-target-like choices in predicative contexts when amplifying high frequency adjectives (see Figure 6). The slopes of both lines indicate that, overall, learners appear to have fewer problems when amplifying high frequency adjectives, as the likelihood of amplifying adjectives in a non-target-like manner decreases as the frequency of adjectives increases.



**Figure 6.** Predicted probability of non-target-like choices in adjective amplification with *very* by adjective frequency and syntactic context

The visualization of the interaction between priming and syntactic context shows that non-target-like choices are significantly less likely to occur in attributive contexts if the context is primed, compared to both in non-primed and primed predicative contexts (Figure 7).

Finally, a significant interaction was reported between syntactic context, gradability, and the emotionality of adjectives. This interaction is visualized in Figure 8 and indicates that in predicative contexts, the probability of making non-target-like choices increases with gradability among negative emotional adjectives, while it remains stable for non-emotional and positive emotional adjectives. In attributive contexts, however, the probability of making non-target-like choices decreases with gradability among positive emotional adjectives, while it remains stable for negative and non-emotional emotional adjectives.

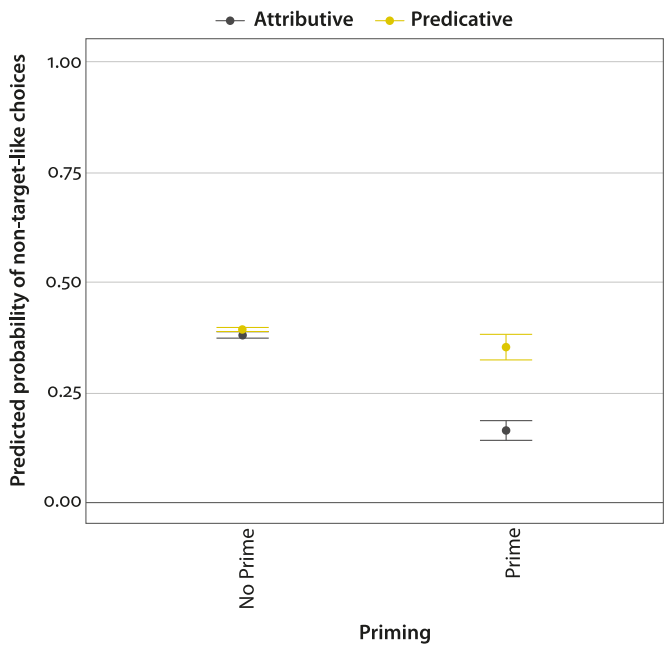


Figure 7. Predicted probability of non-target-like choices in adjective amplification with *very* by priming and syntactic context

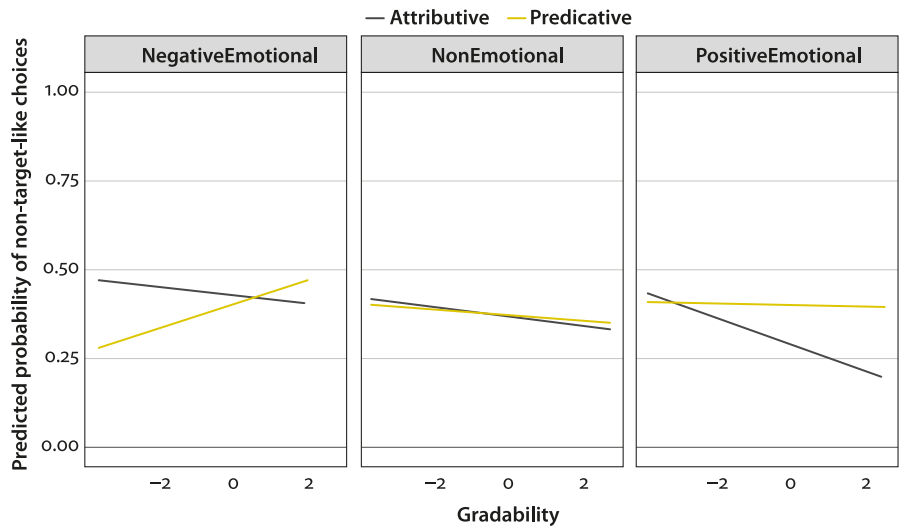


Figure 8. Predicted probability of non-target-like choice against the emotionality of amplified adjectives by syntactic function

## 5. Discussion

The analysis presented here offers insights into how and why NNS from diverse language backgrounds diverge from target-like use of *very* for adjective amplification. The MuPDARF results reveal that, when generalizing over language backgrounds, deviations between NNS and NS are highly adjective-specific with *new*, *dependent*, and *likely* showing the most notable differences between NS and NNS with regard to adjective amplification using *very* (see Figure 3). In contrast, when amplifying *long* and *little*, NNS deviated least from NS. This suggests that NNS have difficulties in acquiring target-like patterns of amplification only when amplifying a rather specific set of adjectives. As such, the MuPDARF analysis has identified a set of adjectives that NNS struggle with. This finding can be implemented by EFL teachers and highlights the importance of teaching collocations rather than words in isolation or amplification without regard to the respective adjective. The present analysis thus substantiates research on adjective amplification which has shifted the focus from the amplifiers themselves to an approach which more thoroughly considers the interdependency of amplifiers and adjectives (see, e.g., Wagner, 2017 or Hendrikx et al., 2019). To exemplify, NS have used *very* with *dependent* only once and have also used *other* amplifiers with *dependent* only once which results in a 1:1 ratio (one instance of *dependent* with *very* against one instance of *dependent* with another amplifier; see [3]). In contrast, NNS have used *other* amplifiers together with *dependent* 37 times but only twice with *very* (see [4] which results in a ratio of 2:37).

- (3) Adjective amplification of *dependent* among NS
  - a. This is when it becomes evident that people are *very dependent* on computers when computers do break down. (ICLE-US-MICH-0007.1)
  - b. Family members are no longer *so dependent* on one another. (ICLE-US-PRB-0039.2)
- (4) Sample of adjective amplification of *dependent* among NNS
  - a. He figured out he was still *greatly dependent* on his parents [...]. (ICLE-BG-SUN-0048.1)
  - b. It is also disquieting that we have become *so dependent* on scientific expertise. (ICLE-SW-LND-0026.4)
  - c. Thus if we look at science and technology in a more global aspect, we will see that our dreams are *entirely dependent* on the modern developments in these fields. (ICLE-BG-SUN-0289.1)
  - d. They were *absolutely dependent* on their memories and on the spoken word. (ICLE-CZ-UNL-0024.1)

Similarly, in the case of *long*, NS have amplified *long* 4 times with *very* and 4 times with *other* amplifiers (ratio 4:4=1), whereas NNS have amplified *long* 19 times with *very* and 27 times with *other* amplifiers (ratio 19:27=0.7). While none of the examples above are unnatural or even unacceptable, the difference in the ratios can give rise to the perception of non-target-likeness or non-nativeness. Lorenz (1999:165), for instance, describes a cumulative effect that causes native-speaking listeners to judge interlanguage as non-native-like: “[y]et what are perceived as no more than ‘stylistic weaknesses’ tend to have an accumulative psychological effect on the reader, leading to an impression of overstatement, crudeness of expression – and ‘foreign-soundingness’”. Non-nativeness in the use of *very* is thus unlikely to result from qualitative differences between NS and NNS, but from subtle but persistently reoccurring instances of statistically unlikely combinations. In other words, non-target-like language use is characterized by the use of combinations of amplifiers and adjectives that occur significantly more, or less, frequently than would be expected given the frequency of these combinations in the speech of L1 speakers.

The importance of the role of the amplified adjective, when it comes to target-like adjective amplification with *very*, is corroborated by the interim analysis that has focused on language- and adjective-specific differences. This analysis found that NNS differ from NS only with respect to a limited set of adjectives (*different*, *difficult*, *important*, and *new*) and has additionally shown that significant differences could only be confirmed for NNS with Spanish, Polish, Swedish, and Italian language backgrounds. The relevance of this finding is that it shows that *very* is not unconditionally overused as was suggested by previous research (see Granger, 1998a; Lorenz, 1999:215; Philip, 2008:1308). Rather, the present study suggests that, in the ICLE data, *very* is only overused with *different* and only by Spanish, Swedish, and Italian NNS. Conversely, *very* is underused when it comes to its co-occurrence with *important* by NNS with a Swedish language background and with *important* and *difficult* by NNS with a Polish language background.

It is important to note that the language background and adjective-specific differences between NS and NNS reported in the interim analysis are not necessarily apparent given the results of the MuPDARF. The reason for this is that the MuPDARF does not aim to detect language-background-specific deviations (which was the aim of the interim analysis) but to identify adjectives that NNS struggle with if we control for language background. Thus, while Polish speakers deviate from target-like amplification of *difficult* and *important*, these adjectives were not identified as being overly problematic in general as, overall, NNS did not deviate from NS when amplifying these two adjectives. Thus, from the broader perspective of MuPDARF, NNS were found to struggle most with the

adjectives *new*, *dependent*, and *likely* while NNS perform target-like when amplifying *long*, and *little*.

The syntactic function of the adjective, while not significant as a main effect, is involved in all three significant interactions that were detected by MuPDARF:

- i. While NNS have fewer difficulties with low frequency adjectives in predicative contexts, they show more non-target-like use of *very* in such contexts when amplifying high frequency adjectives (see Figure 6).
- ii. NNS are less likely to use *very* in a non-target-like manner in attributive contexts if the slot is primed, i.e. if the same amplifier has been used in up to three preceding pre-adjectival slots (Figure 7).
- iii. In predicative contexts, the probability of making non-target-like choices increases with gradability among negative emotional adjectives, while it remains stable for non-emotional and positive emotional adjectives. In attributive contexts, however, the probability of making non-target-like choices decreases with gradability among positive emotional adjectives, while it remains stable for negative and non-emotional adjectives (Figure 8).

The interaction between adjective frequency and syntactic context is particularly relevant from a theoretical perspective because it confirms that the effect of frequency is dependent on the syntactic context in which adjectives occur. The fact that adjective frequency is found to affect non-target-like speech production is, however, rather unsurprising given that frequency information is crucial in both first and second language acquisition (see, for instance, Gass & Mackey, 2002; Tomasello, 2009; or Bybee, 2003). It is, therefore, specifically the conditioning aspect that renders this finding interesting: frequency is not informative by itself in the present context, but only becomes informative for learners once it is transformed into a conditional probability, i.e. the occurrence of something, given something else. This means that raw frequencies by themselves are not meaningful for learners, because learners require additional cues for statistical learning to take effect (see Ellis, 2002; Saffran, Newport, Aslin, Tunick, & Barrueco, 1997; Saffran, 2003). In the present case, the cue is the modified adjective as well as the syntactic context in which an adjective is being used. With this in mind, it appears that the frequency effect is piggybacking on the effect of adjective type, and is additionally conditioned by the syntactic context in which a specific adjective is used.

In order to understand the interaction between syntactic function and priming, it is important to note that attributive contexts exhibit substantially higher variability in terms of the ratio of amplifier types to amplifier tokens: this type-token-ratio is 0.023 in attributive contexts (57/2,451) but only 0.015 in predicative contexts (62/4,261). The interaction between syntactic function and priming can,

therefore, be explained based on the variability that is present in attributive contexts: the effect of priming is surfacing more strongly in attributive contexts because the underlying grammar – and thus the linguistic constraints governing amplifier use in this context – is less stable in NNS due to the higher variability in this context.

The three-way interaction between the syntactic context, gradability, and the emotionality of adjectives supports previous research, which has shown that emotional adjectives are intensified more frequently than non-emotive adjectives (Boucher & Osgood, 1969; Baumeister, Bratslavsky, Finkenauer, & Vohs, 2001). The present findings extend this research by showing under which circumstances emotionality affects adjective amplification. In particular, it shows that neither gradability nor syntactic context matter if the adjective is non-emotional. Exactly why NNS behave more target-like in predicative contexts when amplifying negative emotional adjectives that have low gradability scores, or why NNS behave more target-like when amplifying positive emotional adjectives in attributive contexts, requires further research. It could also be the case that this interaction is merely an artefact that is caused by modeling adjective type as a random effect rather than including it into the fixed effect structure of the model. Alternatively, if the three-way interaction were an artefact, it could be the result of the relatively poor fit of the regression model, which leads us to a necessary discussion of methodological issues.

With respect to methodology, the present study offers relevant insights but also faced noteworthy issues that require further discussion. The random forest model that was fit to the NS data had a remarkably good fit, improved the prediction accuracy by 67.0 percent, and possessed real explanatory power (Harrell's  $C = .847$ , Somers'  $D_{xy} = .694$ ). In contrast, when this model was applied to the NNS data, the fit – while still significant – was remarkably suboptimal (Harrell's  $C = .619$ , Somers'  $D_{xy} = .239$ ) and the accuracy only improved by 15.8 percent when compared to a baseline model. While the poor fit is somewhat disheartening, it is in itself noteworthy as it shows that the underlying grammar of the use of *very* differs substantially between NS and NNS. This is relevant as it indicates that (i) the use of *very* among NNS is influenced by confounding factors that are not relevant in the grammar of NS and (ii) that additional variables should be considered in future research, requiring data that is accompanied by more extensive and thus more detailed information about the learners themselves – including motivational, attitudinal, and psychological aspects – and contextual factors relating to the situational setting. As such, the poor model fit has repercussions for both learner corpus compilation and Learner Corpus Research.

Resulting from the poor model fit of the second random forest model, the mixed-effects model also had only moderate explanatory power and the model fit criteria are below common cut-off values. Somers' Dxy, for example, ranges from -1 to 1, where values around 0 indicate pure chance. A value of merely 0.361 thus shows that the model presented here performs rather poorly. This is substantiated by a low Harrell's C value of 0.681 and the very low  $R^2_{\text{marginal}}$  value of 0.042 (see Table 4 in the Appendix). Harrell's C ranges from 0 to 1 and it has been suggested that models have real predictive capacity with values above .8 (Baayen, 2008: 204). However, while model fit is clearly suboptimal, the model still performs significantly better than chance (L.R.  $\chi^2$ : 500.69,  $df$ =20,  $p$ <.001).

The present study may have profited from alternative operationalizations of linguistic factors, specifically factors that relate to semantic properties of the amplified adjectives. An alternative to the semantic classification used in the present study would have been to follow the classification of adjectives that has been proposed by Paradis (2001, and to a lesser degree 1997). Although promising, the present study has not implemented the operationalization proposed by Paradis (1997, 2001) for two reasons:

1. The classification of adjectives proposed by Paradis would have been collinear with the gradability scores used in the present study;
2. The classification suggested by Paradis (2001) is partly based on the co-occurrence with amplifiers, which would have led to a circularity issue because the classification would have been based on what the analysis aims to explain.

However, implementing a data-driven and collocation-based semantic classification of adjectives would probably have improved the model fit and thus the reliability and explanatory power of the statistical model. Such an approach would also have been relevant theoretically, as it would allow research into adjective amplification to integrate usage-based, bottom-up categorization.

In summary, the results presented here confirm that deviations from target-like language use are notably item-specific with high congruence between NS and NNS achieved for *long* and *little*, while NNS exhibited non-target-like performance for *new*, *dependent*, and *likely*. The results of the present study thus highlight the advantages and draw attention to the necessity of teaching collocations, for example, "by maximizing opportunities to acquire them [collocations] through [...] autonomous implicit learning activities such as reading" (Kennedy, 2003: 467; see also Nesselhauf, 2003). Furthermore, the fact that NNS only struggle with a limited set of adjectives is promising and this can be readily integrated into ELF classrooms. Finally, the analysis presented here calls into question whether over- and underuse can reliably be detected using relative frequencies



rather than conditional probabilities. More generally, this study shows that MuP-DARF and LCR can support EFL teachers by pointing out very specific difficulties and divergences between NS and NNS.

## 6. Conclusion

The present study has shed light on some of the conditioning factors that govern the use of *very* as an adjective amplifier by NNS and offered insights into differences between NS and NNS. In particular, the present study builds upon research which emphasizes the role that adjectives play in amplifier choice (e.g. Wagner, 2017). Furthermore, the finding that NNS struggle with only a rather narrow set of adjectives is promising and can be readily addressed in EFL classrooms. Unfortunately, questions relating to the role of socio-demographic background, differences in the usage profiles of learners, or the impact of motivational factors could not be addressed by the present study and thus remain unresolved. The present study could only rely on the language background of NNS, which is too simplistic to account for the substantial variability among language learners. Future research would therefore profit from incorporating additional information about the background of learners, their motivation, and exposure to the target language into the analysis.

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## Appendix

**Table 4.** Results of the final minimal mixed-effects binomial logistic regression model of the MuPDARF procedure

|                                       | Group(s)  | Variance | Sth. Dev. | L.R. $\chi^2$ | DF      | Significance      |
|---------------------------------------|-----------|----------|-----------|---------------|---------|-------------------|
| Random Effect 1                       | Learner   | 0.03     | 0.16      | 349.08        | 2       | $p < .001^{***}$  |
| Random Effect 2                       | Adjective | 1.26     | 1.12      | 349.08        | 2       | $p < .001^{***}$  |
| Fixed Effect(s)                       | Estimate  | VIF      | OddsRatio | Sth. Error    | z value | Pr(> z )          |
| (Intercept)                           | −1.60     |          | 0.2       | 0.41          | −3.87   | $p < .001^{***}$  |
| LanguageCzech                         | 0.22      | 2.20     | 1.25      | 0.14          | 1.58    | $p = .1148$       |
| LanguageDutch                         | 0.29      | 1.89     | 1.34      | 0.15          | 1.92    | $p = .0543^{(*)}$ |
| LanguageFinnish                       | 0.23      | 1.97     | 1.26      | 0.15          | 1.55    | $p = .122$        |
| LanguageFlemish                       | 0.26      | 1.59     | 1.3       | 0.17          | 1.52    | $p = .1276$       |
| LanguageFrench                        | 0.33      | 2.11     | 1.39      | 0.14          | 2.26    | $p = .0238^*$     |
| LanguageGerman                        | 0.30      | 2.23     | 1.35      | 0.14          | 2.11    | $p = .0347^*$     |
| LanguageItalian                       | 0.09      | 1.97     | 1.09      | 0.15          | 0.6     | $p = .5456$       |
| LanguagePolish                        | 0.39      | 2.41     | 1.48      | 0.14          | 2.86    | $p = .0042^{**}$  |
| LanguageRussian                       | 0.33      | 2.35     | 1.40      | 0.14          | 2.41    | $p = .0161^*$     |
| LanguageSpanish                       | 0.44      | 2.15     | 1.56      | 0.14          | 3.1     | $p = .0019^{**}$  |
| LanguageSwedish                       | 0.52      | 2.47     | 1.67      | 0.14          | 3.8     | $p < .001^{***}$  |
| RelativeFrequency                     | −0.19     | 1.96     | 0.83      | 0.04          | −5.14   | $p < .001^{***}$  |
| SemanticCategoryDimension             | 0.61      | 7.42     | 1.84      | 0.38          | 1.61    | $p = .1073$       |
| SemanticCategoryHumanPropensity       | 0.77      | 15.53    | 2.15      | 0.36          | 2.11    | $p = .0347^*$     |
| SemanticCategoryOther                 | 0.57      | 23.1     | 1.77      | 0.36          | 1.6     | $p = .1087$       |
| SemanticCategoryPhysicalProperty      | 0.27      | 3.54     | 1.31      | 0.41          | 0.65    | $p = .5177$       |
| SemanticCategoryValue                 | 0.24      | 8.92     | 1.27      | 0.37          | 0.65    | $p = .5141$       |
| EmotionalityNonEmotional              | 0.07      | 5.29     | 1.08      | 0.16          | 0.45    | $p = .6554$       |
| EmotionalityPositiveEmotional         | −0.08     | 4.89     | 0.92      | 0.18          | −0.44   | $p = .6565$       |
| PrimingPrime                          | −1.15     | 3.51     | 0.32      | 0.36          | −3.18   | $p = .0015^{**}$  |
| FunctionPredicative                   | −0.03     | 10.11    | 0.97      | 0.19          | −0.15   | $p = .8819$       |
| Gradability                           | −0.08     | 18.78    | 0.92      | 0.07          | −1.12   | $p = .2645$       |
| RelativeFrequency:FunctionPredicative | 0.13      | 3.01     | 1.14      | 0.04          | 3.44    | $p < .001^{***}$  |
| FunctionPredicative:Gradability       | 0.18      | 19.05    | 1.2       | 0.08          | 2.18    | $p = .0295^*$     |

Table 4. (continued)

|                                                               | Group(s) | Variance               | Sth. Dev. | L.R. $\chi^2$ | DF    | Significance     |
|---------------------------------------------------------------|----------|------------------------|-----------|---------------|-------|------------------|
| EmotionalityNonEmotional:FunctionPredicative                  | 0.22     | 8.34                   | 1.24      | 0.19          | 1.14  | $p = .2554$      |
| EmotionalityPositiveEmotional:FunctionPredicative             | 0.62     | 6.06                   | 1.86      | 0.21          | 2.97  | $p = .003^{**}$  |
| PrimingPrime:FunctionPredicative                              | 0.95     | 3.53                   | 2.57      | 0.43          | 2.21  | $p = .0268^*$    |
| EmotionalityNonEmotional:Gradability                          | 0.01     | 14.84                  | 1.01      | 0.08          | 0.13  | $p = .8926$      |
| EmotionalityPositiveEmotional:Gradability                     | -0.01    | 6.8                    | 0.99      | 0.09          | -0.06 | $p = .9529$      |
| EmotionalityNonEmotional:FunctionPredicative:Gradability      | -0.19    | 14.83                  | 0.83      | 0.09          | -2.01 | $p = .0443^*$    |
| EmotionalityPositiveEmotional:FunctionPredicative:Gradability | 0.03     | 6.54                   | 1.03      | 0.11          | 0.32  | $p = .7456$      |
| Model statistics                                              |          |                        |           |               |       | Value            |
| Number of Groups                                              |          |                        |           |               |       | 2,702            |
| Number of cases in model                                      |          |                        |           |               |       | 6,335            |
| Observed misses                                               |          |                        |           |               |       | 3,908            |
| Observed successes                                            |          |                        |           |               |       | 2,427            |
| Residual deviance                                             |          |                        |           |               |       | 7,932.03         |
| $R^2$ (marginal)                                              |          |                        |           |               |       | 0.042            |
| $R^2$ (conditional)                                           |          |                        |           |               |       | 0.315            |
| C                                                             |          |                        |           |               |       | 0.681            |
| Somers' Dxy                                                   |          |                        |           |               |       | 0.361            |
| AIC                                                           |          |                        |           |               |       | 8,000.03         |
| BIC                                                           |          |                        |           |               |       | 8,229.66         |
| Prediction accuracy                                           |          |                        |           |               |       | 65.9%            |
| Model Likelihood Ratio Test                                   |          | L.R. $\chi^2$ : 500.69 |           | DF: 33        |       | $p < .001^{***}$ |

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